

An Attention-based Multi-Scale Feature Learning Network for Multimodal Medical Image Fusion

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1 Introduction

Medical imaging plays an increasingly prominent role in clinical diagnosis. Multimodal medical images can provide information from various aspects, thus helping physicians confirm the diagnosis and make decisions regarding future treatment. For example, Magnetic Resonance Imaging (MRI) images provide the soft-tissue structures of the body; computerized tomography (CT) scans provide bone structure and high-density tissue information; Positron Emission Tomography (PET), and Single-Photon Emission Computed Tomography (SPECT) images can show the metabolic activity of the cells of tissues. To acquire adequate information, physicians must analyze many different medical images, which is time-consuming and laborious. Multimodal medical image fusion (MMIF) can merge the complementary information of original images and present the required information in one fused image. This is clinically significant because physicians can now access detailed information about disease-related changes, thus providing patients with more comprehensive and precise medical treatment.

MMIF methods can be divided into seven categories, including but not limited to multi-scale decomposition, sparse representation, deep learning, etc. [1]. In recent years, many deep learning-based approaches have been proven successful in image fusion because they can extract features from input images and construct a fused image with the information needed. To merge multimodal medical images with more clear edge information and detailed textures, we propose a novel end-to-end framework to merge multimodal medical images with enhanced edges in this paper. Our method combines input images and generate a fused image with as detailed information as possible.

2 Related Work

There are several existing approaches for imaging fusion. Most traditional image fusions are based on the transform domain and are at pixel, feature and decision-level [1]. The usage of fuzzy logic and neural networks with multi-scale decomposition can handle uncertainties and improve efficiency for fusion images. Other iconic works for utilizing traditional image processing algorithms for fusion tasks including Possion Image Editing [2], cross bilateral filter [3], and non-subsampled contourlet transform [4]. Hermessi et al. [5] proposed a CNN-based method in the shearlet domain for extracting feature maps of high-frequency fusion. A multi-generator multi-discriminator conditional generative adversarial network is presented by Huang et al. [6] to fuse functional information and structural information including texture details and dense structure information. This network presents better visual effects and also preserves the approximate maximum amount of information in several MMIF datasets. Fu et al. [7] introduced the residual pyramid attention structure which combines advantages of residual attention and pyramid attention mechanisms in the fusion task. The framework extracts and keeps more information than a single residual attention or pyramid attention mechanism as the number of layers increases and maintains better deep features and expression capabilities. Another network structure on double residual attention [8] combines a residual network and attention to acquire important detailed features while avoiding network gradient vanish or explosion.

3 Materials and Method

3.1 Data

We adapt the commonly used medical image fusion dataset “The Harvard Whole Brain Atlas” for this project. The dataset contains three sets of co-registered multimodal medical image pairs: MRI-CT (184 pairs), MRI-PET (269 pairs) and MRI-SPECT (357 pairs). For now, we will train the model with MRI-CT pairs and extend to the other two if time permits.

3.2 Proposed Method

We introduce a novel end-to-end framework for medical image fusion. Our framework takes a pair of images (I_s, I_f) as input and output a fused image I_o . To be more specific, the framework contains four parts: the edge detector, the feature extractor, the fusion module and finally the reconstruction module. We provide brief explanations of these four parts as follows. Starting with the edge detector, it aims to preserve the boundary information of both input images by using classical edge detection algorithms, i.e., Canny, Sobel, etc. Next, the feature extractor aims to formulate deep semantic

¹Available online at <https://www.med.harvard.edu/aanlib/>

features of both input images, and use as inputs to the fusion module. The architecture of our feature extractor is similar to the one in [7]. The intuition of the fusion module is to determine that for certain pixel in I_o , whether it comes from I_s , I_f or both. The fusion rule can either be $\max(I_s^i, I_f^i)$ for i th pixel or the weighted average of both. Finally, a sequence of convolutional layers are used to reduce channels when reconstructing the final fused image.

Our goal is to coalesce input images to form an output fused image that retain as much details as the input. In other words, the fused image should have a strong correlation with both input images. We argue that the gap between fused image and input images can be measured by the common regression loss function “mean squared error” (MSE). In detail, our loss function is defined as follows:

$$\mathcal{L}(\theta) = \frac{1}{W * H} \left(\sum_{j=1}^M \|I_o - I_j\|_F^2 + \lambda_1 \sum_{j=1}^M \|\nabla I_o - \nabla I_j\|_F^2 \right) + \frac{\lambda_2}{C * W_i * H_i} \sum_{j=1}^M \sum_{k=1}^C \|f_i^k(I_o) - f_i^k(I_j)\|_2^2$$

The first part is a variant of the MSE loss, where M is the number of input images, I_o is the fused image, $\|\cdot\|_F$ is the matrix Frobenius norm. The middle part is to model the fine details of textures, and is realized by the matrix Frobenius norm of the image gradient. The last part is the perceptual loss [9] that models the high-level similarity between two images, where $f_i^k(x)$ is the k th channel in i th layer (with size $W_i \times H_i$) from the pre-trained VGG16 network with input image x , and C is the number of channels. We prefer i to be large, i.e., deeper layer of the VGG network. Finally, θ is the set of network weights to be optimized, λ_1, λ_2 are weight balancing factors of the corresponding loss and the total loss has been normalized by the total pixels in the image with width W and height H .

3.3 Evaluation Metrics

We select some of the most common metrics in the image fusion task. The first one is the Peak Signal to Noise Ratio (PSNR). Since there is no “ground truth” image in the medical image fusion task, we simply compute the PSNR value for (I_o, I_s) and (I_o, I_f) and take the average to obtain the final PSNR value. The other metrics, such as the structural similarity (SSIM), mutual information (MI), and pixel-wise feature MI will be calculated in the same way.

4 Timeline

1. By Nov. 17, we aim to finish the data preprocessing step, complete the model architecture and the loss function
2. From Nov. 18 to Nov. 30, we aim to start training the model (modify the architecture if needed), running experiments (include baseline) and ablation study
3. From Dec. 1 to Dec. 8, we aim to wrap up experiments/results and write our project paper

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